

Dr. KARAN PRATAP SINGH
S/o Shri Prakash Chandra Rathore
(Ph.D. Banaras Hindu University)



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https://www.researchgate.net/profile/Karan_Singh11

FACEBOOK ID:

<https://www.facebook.com/karan.p.singh.14>
PASSPORT NO.:K6082874

Career Objectives

I am looking position as Associate Professor/Research Scientist in the Department of Chemistry, Material Science or Allied subject, where my teaching and research experience will be helpful. To associate with dynamic institute where I will get an opportunity for advancing my knowledge & scientific skills in accordance with latest trends in teaching and research that follows a tradition of anticipating & leading changes.

Ph.D Thesis

Title: Studies on some electrochemical sensor based on composite materials using organically modified sol gel glasses or multiwalled carbon nano tubes.

Supervisor: Prof. Dr. (Mrs) Ida Tiwari, Department of Chemistry, Institute of Science, Banaras Hindu University, Varanasi, Uttar Pradesh, 221005, India.

Publications

1. I. Tiwari, **K.P. Singh**, Manorama Singh, B.C. Upadhyay, V.S. Tripathi, A novel hydrogen peroxide biosensor based on Horseradish Peroxidase incorporated in organically modified sol-gel glass matrix/ graphite paste with multiwalled carbon nanotubes. Analytical Letters, Print ISSN: 0003-2719, Online ISSN: 1532-236X, 2010 vol 43, 2019-2030. **Impact factor: 2.39 (Ph.D. work).**
2. I. Tiwari, **K.P. Singh**, Manorama Singh, An insight review on the application on the polymer carbon nanotubes based composite material in sensor technology, Russian J. Gen. Chem., ISSN 1070-3632, 2009, 79, no.12 PP 2685-2694, Impact Factor : 0.81 (Ph.D. work)

3. I. Tiwari, **K.P. Singh**, In situ synthesis of polymer nanocomposites from PANI/PAA/MWCNTs: Analysis and characterization, International Journal of Polymer Anal. Charact., ISSN 1023-666X print=1563-5341 online 17: 371–380, 2012 , **Impact Factor : 2.58 (Ph.D. work)**
4. I. Tiwari, **K.P. Singh**, Manorama Singh, Polyaniline/ Polyacrylic Acid/ Multi-walled Carbon Nanotube Modified Electrodes for sensing Ascorbic Acid Analytical Methods ISSN 17599 660 2012, 4,1, 118-124 **DOI: 10.1039/C1AY05415E., Impact Factor : 2.896 (Ph.D. work)**
5. I. Tiwari, **K.P. Singh**, Composite materials based on ormosil for the construction of electrochemical sensors and biosensors, Russian Journal of General Chemistry, ISSN 1070-3632, 2012, Volume 82, Number 1, Pages 157-167. **Impact Factor: 0.81 (Ph.D. work)**
6. I. Tiwari, **K.P. Singh**, Fabrication, characterization and application of carbon ceramic nanocomposite prepared by using multiwalled carbon nanotubes and organically modified sol-gel glasses, J. Indian Chem. Soc., Vol. 91, September 2014, pp. 1793-1798. **Impact Factor: 0.447 (Ph.D. work)**
7. V. Gautam, **K.P. Singh**, Vijay Laxmi Yadav, PANI/MWCNTs/Starch Nano-composite Material and Hemoglobin Modified Carbon Paste Electrode for Hydrogen Peroxide and Glucose Biosensing, **International journal of Biological Macromolecules** ISSN 0141-8130 Volume 111, May 2018, Pages 1124-1132, **I.F. 6.953. Corresponding Author (Post Ph.D. work collaborative work)**, (DOI: <https://doi.org/10.1016/j.ijbiomac.2018.01.094>)
8. V. Gautam, **K.P. Singh**, V.L. Yadav, Polyaniline/MWCNTs/Starch modified Carbon Paste Electrode for Non-enzymatic Detection of Cholesterol: Application to Real Sample (Cow Milk), **Analytical and Bioanalytical Chemistry**, **I.F. 4.142** (Accepted 2018), 410, 2173-2181 **Corresponding Author (Post Ph.D. work)**, (DOI: <https://doi.org/10.1007/s00216-018-0880-6>)
9. V. Gautam, **K.P. Singh**, V. L. Yadav, Preparation and Characterization of Green-nano-composite Material based on Polyaniline, Multiwalled Carbon Nano Tubes and Carboxymethyl Cellulose: for Electrochemical Sensor Applications, **Carbohydrate Polymers**, **I.F. 9.381** Volume 189, 1 June 2018, Pages 218-228, <https://doi.org/10.1016/j.carbpol.2018.02.029>, **Corresponding Author (post Ph.D. work)**
10. V. Gautam , **K.P. Singh**, V.L. Yadav, Multicomponent Template Effects-Preparation of Highly Porous Polyaniline Nanorods Using Crude Lemon Juice and Its Application for Selective Detection of Catechol, **ACS Sustainable Chem. Eng.** 2017, 6, 2, 2256–2268, 2017, **I.F. 8.198**, **Corresponding Author (post Ph.D. work)**, <https://doi.org/10.1021/acssuschemeng.7b03705>

11. V. Gautam , A. Srivastava, **K.P. Singh**, V.L. Yadav, Preparation and Characterization of Polyaniline, Multiwall Carbon Nanotubes and Starch Bionanocomposite Material for Potential Bioanalytical Applications, Polymer composite, **IF 3.171 : Corresponding Author (post Ph.D. work)**, ISSN 2728397,38,490-506, 2015
<https://doi.org/10.1002/pc.23608>
12. V. Gautam , A. Srivastava **K.P. Singh**, V.L. Yadav, Vibrational and Gravimetric Analysis of Polyaniline/Polysaccharide Composite Materials, Polymer Science A, ISSN 0965-545X, Polymer Science, Series A, 2016, Vol. 58, No. 2, pp. 206–219. **Impact Factor 0.9 Corresponding Author (post Ph.D. work)**
13. **K.P. Singh**, Preparation of nanocrystalline composite material based on polyaniline, Recent development in material science, Page number 430-437 Research India Press, New Delhi, ISBN 9788189131517, Editor Dr. Sheela Shrivastav, 2011
14. V. Gautam , A. Srivastava **K.P. Singh**, V.L. Yadav, Comparative study of adsorption behavior of Basic Brown-1, Methylene Blue, Methyl Violet on almond shell” recent development in material science, page number 344-355 Research India Press, New Delhi, ISBN 9788189131517, Editor by Dr. Sheela Shrivastav,2011

Educational Details

Class/Course	College/Institute /Department	Board/ University	Year/ Percentage
Ph.D.	Department of Chemistry, Institute of Science	BHU, Varanasi	Degree awarded 3/2/2013
Certificate course in NMR	Proficiency	Indian Institute of Science, Bangalore	2004 B grade
Post Graduate diploma in Chemical Analysis and Quality Management	Directorate of Distance Education	University of Hyderabad, Hyderabad	2003 78%
Certificate training in Chemical analysis and testing	Regional Testing Centre New Delhi	Ministry of Small Scale Industry, Government of India	2001
M.Sc.	Bareilly College Bareilly	MJPRU Bareilly	2000 57.1%
B.Sc.	Bareilly College Bareilly	MJPRU Bareilly	1997 53%
Intermediate	Kendriya Vidyalaya (Cooch Bihar)	CBSE	1994 63.2%

High School	Kendriya Vidyalaya-1 (JAT Centre, Bareilly)	CBSE	1992 52.8%
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Work Experience

From 2000 – till date

1. **Associate Professor**, Department of Chemistry, OPJS, University 1 Feb – 10 Sep 21
2. **Assistant Professor**, Department of Chemistry JJTU, Jhunjhunu, Raj. 2018- 15 January 2021
3. **Assistant Professor** (Adjunct), Department of Chemistry, JJTU, Jhunjhunu, Rajasthan 2015-2018
4. **Mentor of Change**, Atal Incubation Lab, Sacred Heart School, Bareilly, Nominated by Niti Ayog, Govt. of India.
5. **Assistant Professor** MJPRU Bareilly. 2015-2016
6. **Founder President**, Mahamana Innovative Technologies Welfare Society, Nawabganj Bareilly. 2012- Till date
7. **Lecturer**, MD Senior Secondary School, Gurgaon. 2013-2014
8. **Assistant Professor**, Invertis University, Bareilly. 2014-2015
9. **Chemist**, Metal core private limited, Bangalore 2005-2006
10. **Copy Writer** Sainath Enterprises, Bangalore 2004
11. **Chemist**, Esdee paints limited, Ahmadabad. 2002-2003
12. **Chemist**, Moon light paints industry, Amritsar. 2001-2002.

Achievements and skills

1. **Develop a low cost portable potentiostat system (electrochemical interface)** for the fabrication of sensor and biosensor.
2. **Best poster award for Presenting research paper entitled** “Crystallographic information of nanocrystalline composite material (NCCM) based on polyaniline: Electron Diffraction Study” at “National conference on advance and futuristic trend in Material Science (AFTMS 2011) March 26-27, 2011, organized by Department of Physics, MJP Rohilkhand University, Bareilly.
3. **Best poster award (coauthor)** for Presenting research paper entitled “Application of low cost adsorbent (treated peanut hull) for removal of colors pollutants (Methylene Blue and Methyl Violet) from waste water effluents” at National Seminar on Water & Environment Challenges for water resource management in 21st Century dated 12-13 march 2011, organized by MVM College, Bhopal.
4. **Achieved best Prize in poster presentation (coauthor)** for research paper “**FTIR analysis of Polyaniline and polysaccharide based composite**” International Conference on “Harnessing Engineering, Technology, and Innovation for sustainable Development” Dr. S.S. Bhatnagar University Institute of Chemical Engineering and Technology, Punjab University, Chandigarh, India on 19-20th September 2014
5. **Achieved best poster Prize in poster presentation (coauthor)** for research paper entitled “**Investigation on “multi-component composite as an improved adsorbent”** at International Conference on Nanoscience and Nanotechnology organized by Department of

Applied Physics, School for Physical Sciences at Babasaheb Bhimrao Ambedkar University Lucknow on 18-20 November-2013.

6. **Best Poster Award (coauthor)** for poster presentation of research paper entitled “**Polyaniline/MWCNTs/Starch : Hemoglobin based Hydrogen Peroxide Biosensor**” at institute day of IIT-BHU
7. Qualified GATE-2005 exam.
8. Have knowledge of basic computers and hand on experience on different analytical instruments.
9. In hand practical experience on testing and chemical analysis of different industrial/commercial products as per protocols referred by IS, ASTM and BIS of various materials such as water, paint, cement, tooth-paste, soap, detergent, pesticides, stainless steel, alloy, oil, bitumen and many others.
10. Participated in co-curricular activities at college and university level.
11. Get position in university youth festival BHU Varanasi **SPANDAN 2009** in “Instrumental solo”; Participate and secure **First prize** in “Instrumental solo” “**AAKANSHA 2010**” Faculty of Science youth fest. BHU; Participate and secure **Second prize** in “Instrumental solo” at “**AAKANSHA 2009**” faculty of science youth fest. BHU Varanasi. Participate and secure **Third prize** in “Instrumental solo” at “**AAKANSHA 2008**” faculty of science youth fest. BHU Varanasi.
12. Sponsor, execute and coordinate social activities carried under the programs of MITWS; Establish public library Mahamana Charitable Book Bank- for poor and needy Students; EDICOR program to support young researchers; Awareness about non renewable source of energy under URJAA; Mentor for science club “ Mahamana Vigyaan Manch”.

Research Interest

Favorite subjects: Electrochemistry, Nanotechnology, Sol-gel science, Polymer science, Intrinsic conducting polymers, Colloidal Chemistry.

Research area: Sensor and Biosensors Technology, Ormosil, Composite materials, Nano-materials and Nano Technology, Conducting polymers, Low cost Adsorbents.

Conference attendant and presented papers

1. Poster Presentation on the research paper entitled “Powder XRD Analysis of PANI/PAA/MWCNTs based composite materials prepared in two different acids”, national symposium on organic synthesis and advance material, 2-3 March 2014, organized by Department of Chemistry, BHU- Varanasi.
2. Poster Presentation of research paper entitled “Development and characterization of an Ormosil-MWCNTs composite” at India Japan workshop on biomolecular electronic and organic nanotechnology and environment protection 2009 dated 17-20 Dec 2009, organized by NPL, New Delhi.
3. Oral Presentation of research paper entitled “Investigation of Polyaniline Having Single and Dual Dopants” at “national conference in advance polymer science and technology 2010” dated 22-24 Oct 2010, organized by NIT, Hamirpur.

4. Oral Presentation of research paper entitled "Composite material of polyaniline along with polyacrylic acid and MWCNTs" at national conference recent trend exotic material 2010" dated 26-28 Aug-2010 organized by Sharda University, Noida.
5. Oral Presentation of research paper entitled "Application of Ganga sand/polyaniline composite material as low cost adsorbent for the removal of methyl violet " at NSAFM-2012" dated 11-12 Feb-2012 organized by Department of Chemistry, BHU- Varanasi.
6. Oral Presentation of research paper entitled "Development of microprocessor based low cost portable potentiostat device for amperometric sensor/biosensor application" at NSAFM-2012" dated 11-12 Feb-2012 organized by Department of Chemistry, BHU-Varanasi.
7. Oral Presentation of research paper entitled "Characterization techniques for composite materials, nano materials and their application for electrochemical sensors" at National conference "Current concept and frontier advances in science educational research" March 5-6. 2011, organized by TD College, Jounpur UP.
8. Poster Presentation of research paper entitled "Preparation of nanocrystalline composite material based on polyaniline" at "RDMS-2011" dated 11-12 Feb-2011, organized by Feroz Gandhi College, Rai Bareli.
9. Poster Presentation of research paper entitled "Removal of dye from waste water using low cost adsorbent" at National seminar, "Impact of environmental changes of human life" dated 20-21 Nov 2010, organized by SSK Girls degree college, Allahabad.
10. Participated in 2 days international conference 3rd Bangalore Nano dated 8-9 Dec 2010, organized by JNCSR, Bangalore.
11. Participated in 2 days „National symposium on current trends in chemistry“, dated 24-25 March 2007, organized by Department of Chemistry, BHU- Varanasi.
12. Participated in 10th CRSI national symposium in Chemistry, dated 1-3 Feb 2008, organized IISC, Bangalore.
13. Participated in 2 days workshop entitled "Paints and allied product testing and analysis" at BIS central laboratory, Sahibabad.
14. Participated in 2 days workshop entitled "National workshop on Jugaad in Manufacturing" dated 8-9 Jan 2011, organized by Main workshop, Institute of Technology, BHU- Varanasi.
15. Participated in Recent trends in Chemistry, dated 19-20 Feb 2011, Organized by Department of Chemistry, BHU-Varanasi.
16. Poster Presentation of research paper entitled "Application of low cost adsorbent (treated peanut hull) for removal of colors pollutants (Methylene Blue and Methyl Violet) from waste water effluents" at National Seminar on Water & Environment Challenges for water resource management in 21st Century dated 12-13 march 2011, organized by MVM College, Bhopal.
17. Poster Presentation of research paper entitled "Comparative study on the adsorption behavior of some basic dyes from waste water using activated sand (from Ganga River)" at Annual International Conference on Water & Wastewater Management "H2O World 2011" 11 March, At Hotel Eros international, New Delhi.
18. Poster Presentation of research paper entitled "Removal of Brilliant Blue from dye wastewater uses activated sand (from Ganga River) by adsorption" at National conference "Current concept and frontier advances in science educational research" March 5-6. 2011, organized by TD College, Jounpur UP.

19. Poster Presentation of research paper entitled “Crystallographic information of nanocrystalline composite material (NCCM) based on polyaniline: Electron Diffraction Study” at “National conference on advance and futuristic trend in Material Science (AFTMS 2011) March 26-27, 2011, organized by Department of Physics, MJP Rohilkhand University Bareilly.
20. Poster Presentation of research paper entitled “Ganga sand used for removal of dye (methylene blue and methyl violet) by adsorption method” at “National conference on advance and futuristic trend in Material Science (AFTMS 2011) March 26-27, 2011, organized by Department of Physics, MJP Rohilkhand University Bareilly.
21. Poster Presentation of research paper entitled “Implementation of Biosensors in Food Industry for Detecting Unwanted Threats: A Necessary Safety Measure” at “National conference on recent advances in the development of fermented food (RADFF-2011) April 8-9, 2011, organized by Centre for Food Science and Technology, Institute of Agriculture Science, BHU, Varanasi.
22. Poster Presentation of research paper entitled “Removal of Victoria blue by using Ganga river sand as a low cost adsorbent” at international conference “CONIAPS-XIII 13th Conference of International Academy of Physical Sciences”, June 14-16, 2011, organized by University of Petroleum and Energy Studies, Dehradun.
23. Poster presentation (coauthor) of research paper entitled “Commercially available water based adhesives: a comparative studies” International Conference on “Harnessing Engineering, Technology, and Innovation for sustainable Development” Dr. S.S. Bhatnagar University Institute of Chemical Engineering and Technology, Punjab University, Chandigarh, India on 19-20th September 2014.
24. Poster presentation (coauthor) of research paper entitled “FTIR analysis of Polyaniline and polysaccharide based composite” International Conference on “Harnessing Engineering, Technology, and Innovation for sustainable Development” Dr. S.S. Bhatnagar University Institute of Chemical Engineering and Technology, Punjab University, Chandigarh, India on 19-20th September 2014.
25. Poster presentation of research paper entitled “ORMOSIL/MWCNTs based bio-nano-composite electrode material” in International Conference on “Harnessing Engineering, Technology, and Innovation for sustainable Development” Dr. S.S. Bhatnagar University Institute of Chemical Engineering and Technology, Punjab University, Chandigarh, India on 19-20th September 2014.
26. Oral presentation (coauthor) of research paper entitled “Preparation and Characterization of Polyaniline/Kaolin Composite Material” in national conference -IRMAST at Rajshree College, Bareilly.
27. Oral presentation of research paper entitled “Application of Polyaniline based Composite Material for Biosensors” in national conference -IRMAST at Rajshree College, Bareilly.
28. Oral Presentation of research Paper entitled “Polyaniline/MWCNTs/Starch Composite Material for Hydrogen Peroxide Sensor” in International Conference on Recent Trends in Engineering and Materials Science (ICEMS-2016) dated March 17-19, 2016 at Jaipur National University, Jaipur.
29. Participate on National Workshop on Innovative Research on Materials in Science & Engineering [IRMSE-16], under Faculty Development Program (FDP) (17 – 22 April, 2016) at MJPRU, Bareilly.

30. Participate on National Workshop on Innovative and Research trends in mechanical and production engineering, under Faculty Development Program (FDP) (25 – 29 April, 2016) at MJPRU, Bareilly.
31. Participate on one week workshop on “Simulation of wireless Sensor network” 2-8 may 2016, at MJPRU, Bareilly.

Personal details

Father:	Retired Subedar Major Shri Prakash Chandra Rathore
Mother:	Smt. Suman Lata Rathore (house wife)
Brothers:	R.P.S. Rathore and M.P. Singh (Work as scientists in DRDO); DPS Rathore (Software Engineer)
Marital status:	Single
Physical status:	Physically challenged; post polio permanent paralysis in left leg.
Date of Birth:	7-6-1976
Category:	GENERAL, CAST
Hobbies:	Playing and compose music, poetry, travelling and exploration of natural phenomena, wild life and archeological sites, internet browsing
Strength:	Communicating; Flexibility and Adaptability; Quick learner; Stress tolerance; Decision making skills; Teaching, Creating Ideas: Creativity, Kind hearted and Honest
Weakness:	I focus too much on the details; I have a hard time letting go of a project; I have trouble saying “no.”; I get impatient when projects run beyond the deadline; I could use more experience; I sometimes lack confidence.

References

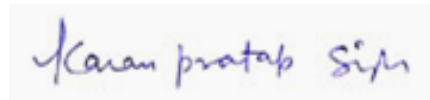
1. Prof. Ida Tiwari, Department of Chemistry, Institute of Science, Banaras Hindu University, Varanasi, Uttar Pradesh, India idatiwari_2001@rediffmail.com , +919415813020
2. Prof. Anchal Srivastava, Department of Physics, Institute of Science, Banaras Hindu University, Varanasi, Uttar Pradesh, India. anchalbhu@gmail.com, +919453203122
3. Prof. Vijay Laxmi Yadav, Department of Chemical Engineering, Indian Institute of Technology, Varanasi, Uttar Pradesh, India, vlyadav.che@itbhu.ac.in, +919935048388
4. Prof. Uma Sankar Rai, Department of Chemistry, Institute of Science, Banaras Hindu University, Varanasi, Uttar Pradesh, India, usrai_bhu@yahoo.co.in, +918887727309
5. Prof. R.C. Tiwari (Department of Agronomy, CEO and Advisor to Chancellor/Chairperson SJJT University), drtiwarirc@yahoo.co.in, +919351418100
6. Prof. R.A. Singh, Department of Chemistry, Institute of Science, Banaras Hindu University, Varanasi, Uttar Pradesh, India, prrasingh@gmail.com, +918840013433

Declaration

I hereby declare that the information furnished above is true to the best of my knowledge. I do hereby declare that above particulars of information and facts stated are true, correct and complete to the best of my knowledge and belief.

Place: Nawabganj, Bareilly

Date:...../...../20...



Dr. Karan Pratap Singh